

Nafis Ahtasum

COMPUTATIONAL MATERIALS RESEARCHER · RESEARCH AND ANALYSIS INSTITUTE (RAI)

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SUMMARY

Computational Materials Researcher at CMCE (RAI), developing computational methods that integrate **data-driven inverse design**, **physics-informed machine learning**, **Density Functional Theory (DFT)**, and **high-throughput screening** to accelerate materials design and discovery for energy systems and catalysis, enabling new classes of *perovskites*, *functional polymers*, and *nanomaterials*; founder of **RAI** and **Resogram**, initiatives dedicated to open infrastructure, democratizing research access, and facilitating global scientific collaboration.

EDUCATION

Bangladesh University of Textiles (BUTEX)

B.Sc. in Textile Engineering (Major in Apparel Engineering)

Dhaka, Bangladesh

Jan. 2019 – Aug. 2026

- CGPA: 2.88/4.00
- **Undergraduate Thesis:** *Comparative Analysis of Microfiber (Microplastic) Release from Industrial Garment Washing Techniques Using Machine Learning and Optical Image Processing.*
- **Relevant Coursework:** Engineering Mathematics, Computational Modeling, Physical Chemistry, Polymer Science, Fiber Physics, Statistical Quality Control.

PUBLICATIONS (Full list on [Google Scholar](#))

- **Ahtasum, N.**, Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., and Roni, M. R. A., 2026. **Backward mapping from target optoelectronic phenotypes to chemical genomes for the discovery of lead-free double perovskites.** *Materials Today Physics* (Q1, Impact Factor: 9.3), 66, p.102180.
- **Ahtasum, N.**, Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., Rashel, M. R., Jamil, H., Islam, A. K., and Tlemcani, M., 2026. **Genome-Guided Interpretable Screening of Phase-Stable, Lead-Free Double Perovskite Absorbers for All-Inorganic Semiconductors, Sensors, and Photovoltaics with DFT-Validated Design Rules.** *arXiv Preprint*, arXiv:2605.22887.

RESEARCH EXPERIENCE

Computational Materials Researcher, Center for Material, Climate and Energy (CMCE), RAI

2024 – Present

Supervisor: *Dr. M. M. Harussani*

Dhaka, Bangladesh

- **Backward mapping from target optoelectronic phenotypes to chemical genomes for the discovery of lead-free double perovskites.** *Materials Today Physics* (Q1, Impact Factor: 9.3), 66, p.102180.
Screened 13,088 $A_2BB'X_6$ double perovskites via EA-optimized tree ensembles ($R^2 = 0.93$) and custom high-recall loss functions to preserve synthetically accessible metastable phases ($\Delta E_{\text{hull}} \leq 25$ meV/atom).
- Applied SHAP interpretability to reduce candidate search space by 99.9% and DFT-validated 7 stable endpoints; confirmed 90
- **Title: First-Principles Adsorption Dynamics and Stimuli-Responsive Drug Delivery on 2D MXenes** [Under Manuscript Preparation]
Investigated adsorption mechanisms and pH-responsive release kinetics of capecitabine on Ti_3C_2 MXene via DFT and MD simulations; quantified strong exothermic binding ($E_{\text{ads}} \approx -3.2$ eV) stabilized by charge transfer ($0.3\text{--}0.4 |e|$) and demonstrated tumor-acidic release kinetics ($\Delta E_{\text{rel}} \approx +3.2$ eV).
- **Title: High-Throughput DFT Screening of Deep Generative Candidates for HER-Suppression** [Under Manuscript Preparation]
Conducted Density Functional Theory (DFT) calculations on metallic candidates generated via a Deep Generative Framework ($> 5,000$ database compounds, 400k structures), isolating 18 HER-suppressing Co-Zr-X ($X = \text{Ba, Ti}$) trimetallics for cathodic glycerol electroreduction.

Research Collaborator, Department of Computer Science, University of Idaho, USA

June 2026 – Present

Supervisor: *Prof. Hasan Jamil*

Remote, USA

- **Title: ML Surrogate-Driven Inverse-Design Pipeline Development for Perovskite Halide and Oxides** [Ongoing]
Developing inverse-design computational workflows, curating benchmark materials datasets, and training ML surrogate

models to screen phase-stable perovskite oxide candidates for optoelectronic applications.

- **Genome-Guided Interpretable Screening of Phase-Stable, Lead-Free Double Perovskite Absorbers for All-Inorganic Semiconductors, Sensors, and Photovoltaics with DFT-Validated Design Rules**

Deployed a genome-guided, physics-informed ML framework across 3 descriptor families with staged constraint filtering over 13,000 compositions, isolating 5 DFT-validated, strictly stable ($E_{\text{hull}} \leq 0$ meV/atom) optoelectronic semiconductors.

Research Collaborator, Universidade de Évora, Portugal

Jan. 2025 – Apr. 2026

Supervisors: Prof. Dr. Mouhaydine Tlemcani and Dr. Masud Rana Rashel

Remote, Portugal

- **Title: Accelerated discovery of stable lead-free halide double perovskites for optoelectronics via evolutionary-tuned machine learning and density functional theory** **[Under Manuscript Preparation]**

Designed end-to-end computational screening workflows, curated materials databases, and developed predictive machine learning pipelines for DFT-validated lead-free double perovskite discovery.

Undergraduate Researcher, Bangladesh University of Textiles (BUTEX)

Jan. 2024 – May 2025

Supervisors: Most. Jesmin Nahar Kamona and Nusrat Jahan

Dhaka, Bangladesh

- **Title: Experimental Quantification and Microscopic Tracking of Microfiber Shedding Dynamics**

Led experimental filtration, gravimetric analysis, and material characterization via FE-SEM/EDS, FTIR-ATR, and ImageJ optical microscopy; presented findings at the *International Conference on Textile Science and Engineering 2025*.

- **Title: Hydraulic Hot-Press Formulation and Mechanical Characterization of Recycled Waste Composites**

Formulated compression-molded composite panels from garment offcuts and polypropylene (PP) matrices via a CARVER hydraulic press; mechanical UTM testing outperformed ASTM D350 OSB/plywood benchmarks (*JFPC*, 2024; *ICONFIB 2024*).

ADDITIONAL RESEARCH EXPERIENCE

Lead Research Fellow, V-Trion Textile Research GmbH, Austria

July 2025 – Present **[Ongoing]**

Supervisor: Dr. Ashaduzzaman Khan

Remote, Austria

- Directing and mentoring 6 remote research teams (28 undergraduate/graduate students), supervising 6 critical review papers on silicon-anode binders, solid polymer electrolytes, wearable biosensors, wound-infection tracking, biodegradable DAC sorbents, and electrochemical CO₂ conversion pathways.

Science Policy & Systems Researcher, PIRI-BD Initiative

Apr. 2025 – Present

[Under Manuscript Preparation]

Supervisors: Md. Rafiul Alam Roni and Prof. A. K. M. Kamrul Islam

Dhaka, Bangladesh

- Co-authoring the strategic vision manuscript "*Towards the Establishment of a Private Interdisciplinary Research Institute*" alongside international faculty and students (NC A&T, Univ. of Iowa, Univ. of Maine, Univ. of Évora), formulating operational blueprints for research governance, funding models, and infrastructure.

PROJECTS

Discovery-of-perovskite-halide- (Materials Informatics, DFT)

 [Repository](#)

[Jupyter Notebook, Python] Developed a machine learning and materials informatics pipeline to discover novel lead-free $A_2BB'X_6$ double perovskites with targeted thermodynamic stability and Shockley–Queisser limit band gaps using evolutionary-optimized surrogates and DFT validation.

double-perovskite-Oxide (Materials Informatics, DFT)

 [Repository](#)

[Jupyter Notebook, Python] Built a multi-objective machine learning and screening pipeline trained on > 5,000 double perovskite oxide ($A_2BB'O_6$) datasets to resolve thermodynamic phase stability, metal/insulator separation, and optimal bandgap windows.

genome-guided-perovskite (Data-Driven Discovery, ML, DFT)

 [Repository](#)

[Jupyter Notebook, Python] Engineered a high-recall machine learning screening framework trained on 1,221 DFT calculations to evaluate 13,088 lead-free compositions across 13 descriptors, identifying 5 DFT-confirmed phase-stable semiconductors ($\Delta E_{\text{hull}} \leq 0$ meV/atom).

TECHNICAL & COMPUTATIONAL SKILLS

Materials & Polymer Informatics / ML:	Material-discovery architectures, evolutionary algorithms, tree ensembles, neural networks, and physics-informed ML surrogates for high-throughput screening, inverse design, and property prediction using Scikit-learn, Pandas, NumPy, Matplotlib, and Seaborn.
First-Principles, Atomistic & Catalysis:	Density Functional Theory (DFT) and Molecular Dynamics (MD) via CASTEP, DMol3, Quantum ESPRESSO, VASP, Gaussian 09W, Materials Studio, and VESTA for modeling band structures, charge dynamics, adsorption, and electrocatalytic pathways (CO ₂ RR).
Experimental Characterization:	FE-SEM/EDS, FTIR-ATR, optical microscopy with ImageJ particle tracking, Universal Testing Machines (UTM), and hydraulic hot-press fabrication (CARVER press) for composite and polymer systems.
Software, Infrastructure & Visualization:	Publication-standard plotting and design via OriginPro, Adobe Illustrator, and Canva, alongside Linux/Bash environments, Git/GitHub version control, VS Code, L ^A T _E X, Mendeley, and Zotero.

OPEN-SCIENCE PLATFORMS & LEADERSHIP

Resogram (<i>Founder & Product Lead</i>)	2025–Present
Launched an academic network connecting 1,001 active researchers globally with 91 publications/posts , directing feature architecture for interdisciplinary matching and open peer review.	
Research and Analysis Institute (RAI) (<i>Founder & Managing Director</i>)	2024–Present
Established an open-access institute providing computational materials training, HPC access, and mentorship, alongside global seminar series (<i>Aspire Talks</i> , <i>RAI Research Podcast</i>) featuring leads from Google, Meta, and Tesla.	

TEACHING, SUPERVISION & MENTORING

Training Programs & Technical Workshops (RAI)	2024–2025
<i>Instructor / Co-Instructor:</i> Literature Navigation and Critical Analysis (44 students, Oct. 2025); AI in Scientific Research (32 students, Sep. 2025); 7-Day Hands-on DFT Simulation Training (56 students, Nov. 2024); Elite Software Training for Research: Quantum ESPRESSO, Gaussian, Materials Studio (122 students, Oct. 2024).	
Research Supervision (RAI Next Generation Researchers Initiative)	2025–2026
<i>Co-Supervisor (DFT):</i> Point Defect Kinetics & Exciton Dynamics in Co-Doped Cs ₂ AgBiCl ₆ ; First-Principles Adsorption Dynamics of H ₂ SO ₄ on MXene Monolayers.	

PROFESSIONAL EXPERIENCE

Computational Materials Researcher , Center for Material, Climate and Energy (CMCE), RAI	2024–Present
Managing Director , Research and Analysis Institute (RAI), Dhaka-1208, Bangladesh	2026–Present
Research Assistant , UniversaPulsar Lda (Spin-off of University of Évora), Évora, Portugal	2025–2026
Industrial Trainee , Knit Concern Ltd., Narayanganj, Bangladesh	2025

GLOBAL OUTREACH & TECHNICAL SEMINARS

Program Director & Executive Producer , RAI Global Lecture Series	2025
Organized and hosted featured technical lectures with international scholars and industry leads:	
• MIT Media Lab: <i>Innovations in Piezoelectricity & Bioengineering</i> (Ikra Iftekhar Shuvo, MIT)	
• Industry Keynotes: Featured speakers from Google (Shahed Ahmed), Meta (Iqbal Hossain), and Tesla (Alan Kabir)	
• European Research Institutes: Lectures with CNRS France, Univ. of Liège, Univ. of Helsinki, and V-Trion Austria	

ACADEMIC REFERENCES

Most. Jesmin Nahar Kamona	<i>Dhaka, Bangladesh</i>
<i>Assistant Professor, Department of Apparel Engineering, Bangladesh University of Textiles</i> Email: jasmine@butex.edu.bd	
Prof. Hasan Jamil, Ph.D.	<i>Moscow, ID, USA</i>
<i>Associate Professor, Department of Computer Science, University of Idaho</i> Email: jamil@uidaho.edu	
Dr. M. M. Harussani, Ph.D.	<i>Kuala Lumpur, Malaysia</i>
<i>Principal Investigator, CMCE, RAI Postdoctoral Fellow, Universiti Malaya</i> Email: mmharussani@um.edu.my	